

Retail Sales & Profitability Analysis

Business Analyst Portfolio Project

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Tools Used: Google Sheets | SQL (BigQuery) | Tableau

Project Type: Retail Sales & Profitability Analysis

1. Executive Summary

This project analyses retail sales and profitability using the Sample Superstore dataset to identify the key factors influencing commercial performance. The analysis focuses on sales, profit, product categories and sub-categories, customer segments, discounting, product-level performance and changes in performance over time. Google Sheets was used for data preparation and quality checks, Google BigQuery was used for SQL-based analysis, and Tableau Public was used to develop an interactive dashboard.

The analysis identified important differences between sales performance and profitability across the business. While some categories, customer segments and products generated strong sales, revenue alone did not always translate into strong profit. Discount analysis also highlighted the importance of examining pricing decisions alongside profitability, while product-level analysis helped identify areas of stronger and weaker commercial performance.

Overall, the project demonstrates how transactional retail data can be transformed into actionable business insights to support profitability management, product strategy, customer analysis and data-driven decision-making.

2. Introduction

Retail businesses generate large volumes of transactional data relating to customers, products, sales, discounts and profitability. Analysing this information can help organisations understand which areas of the business generate value, identify underperforming products and evaluate whether sales growth is translating into sustainable profit.

This project analyses the Sample Superstore retail dataset to evaluate sales and profitability performance across product categories, sub-categories, customer segments and individual products. The analysis also examines yearly performance and the relationship between discounting and profitability to identify potential commercial risks and opportunities.

The project follows an end-to-end business analytics workflow, beginning with data preparation and quality checks in Google Sheets, followed by SQL analysis using Google BigQuery and interactive data visualisation in Tableau Public. The resulting analysis converts transactional retail data into business-focused insights that can support product, pricing and profitability decisions.

3. Business Problem

Strong sales performance does not necessarily result in strong profitability. A retail business may generate substantial revenue from particular products, categories or customer segments while experiencing low or negative profit because of discounting, product-level performance and differences in profit margins.

The business problem addressed in this project is how sales and profit performance vary across the Superstore business and which factors contribute to stronger or weaker profitability. The analysis therefore examines category and sub-category performance, customer segments, yearly trends, discounting and individual products to identify where revenue is being converted effectively into profit and where profitability may be under pressure.

4. Project Aim and Objectives

4.1 Project Aim

The aim of this project is to analyse retail sales and profitability performance using the Sample Superstore dataset and identify the key factors influencing commercial outcomes. The project aims to evaluate performance across product categories, sub-categories, customer segments, discounts and individual products, while using SQL and data visualisation to translate transactional data into actionable business insights.

4.2 Project Objectives

- Analyse overall sales and profitability performance.
- Compare sales and profit across product categories and sub-categories.
- Evaluate customer segment performance to identify differences in commercial contribution.
- Analyse yearly trends to understand changes in sales and profitability over time.
- Examine the relationship between discount levels and profitability.
- Identify high-performing and loss-making products.
- Use Google BigQuery and SQL to perform structured retail analysis.
- Develop an interactive Tableau dashboard to communicate key findings visually.

- Translate analytical findings into practical recommendations for improving retail profitability.

5. Dataset Overview

The project uses the Sample Superstore retail dataset, which contains transactional information relating to customer orders, products, sales, discounts and profitability. Each record represents an individual product line within a customer order, allowing performance to be analysed across multiple dimensions of the retail business.

The dataset includes variables such as order and shipping dates, customer segment, product category and sub-category, sales, quantity, discount and profit. These fields enable both financial and operational performance to be examined from different perspectives.

The analysis primarily focuses on **Sales** and **Profit** as the core performance measures, with **Category, Sub-Category, Segment, Product, Year and Discount** used to investigate the factors associated with differences in commercial performance.

Dataset Summary

Attribute	Details
Data Source	Sample Superstore Retail Dataset
Data Type	Transactional Retail Data
Core Measures	Sales, Profit, Quantity, Discount
Key Dimensions	Category, Sub-Category, Segment, Product, Year
Analysis Focus	Sales Performance and Profitability
Data Preparation Tool	Google Sheets
Final Dashboard Tool	Tableau Public

6. Data Preparation and Cleaning

6.1 Data Preparation in Google Sheets

The Sample Superstore dataset was imported into Google Sheets for initial preparation and validation before SQL analysis. The dataset was reviewed to understand its structure and ensure that key fields required for the analysis—including sales, profit, discount, customer segment, category, sub-category and product information—were suitable for further analysis.

A cleaned working dataset was prepared so that the same underlying data could be used consistently across Google Sheets, Google BigQuery and Tableau.

6.2 Data Quality Checks

Data quality checks were performed to identify potential issues that could affect the reliability of the analysis. Key fields were reviewed for missing values, duplicate records, inconsistent formatting and appropriate data types. Sales, profit, quantity and discount fields were checked as numerical measures, while categorical fields such as Segment, Category and Sub-Category were reviewed for consistency.

Date fields were also reviewed and prepared for time-based analysis. A year field was derived from the relevant date information to support comparison of sales and profitability performance over time.

6.3 Preliminary Analysis

Google Sheets was also used for preliminary exploration of the cleaned data. Pivot tables and summary calculations were used to review performance across key dimensions and validate patterns before progressing to SQL analysis. This provided an initial understanding of the dataset and helped define the analytical questions explored further in BigQuery and Tableau.

6.4 Final Analytical Structure

The prepared dataset retained the transactional structure required for detailed retail analysis while ensuring that the key measures and dimensions were consistently formatted. This structure supported analysis of overall performance, categories and sub-categories, customer segments, yearly trends, discount behaviour and individual product profitability.

7. SQL Analysis in Google BigQuery

7.1 SQL Analysis Approach

Following data preparation, the cleaned retail dataset was imported into Google BigQuery for structured SQL analysis. SQL was used to calculate key performance measures, compare performance across different areas of the business and investigate factors associated with profitability.

The analysis progressed from overall performance measures to more detailed comparisons across customer segments, product categories and sub-categories, yearly trends, discount levels and individual products. This approach enabled both high-level business performance and specific areas of profitability risk to be examined.

7.2 Key SQL Analysis Performed

- Calculated overall sales and profit KPIs to establish total business performance.
- Compared sales and profitability across customer segments.
- Analysed category and sub-category performance to identify stronger and weaker areas of the product portfolio.
- Examined yearly sales and profit trends to assess changes in performance over time.
- Analysed discount levels alongside profitability to investigate whether heavier discounting was associated with weaker commercial outcomes.
- Identified high-performing and loss-making products through product-level profitability analysis.
- Ranked analytical outputs to highlight areas requiring greater business attention.

7.3 SQL Skills Demonstrated

The BigQuery analysis demonstrated practical use of SQL techniques including **SELECT**, **SUM**, **AVG**, **COUNT**, **GROUP BY**, **ORDER BY** and filtering conditions. These techniques were applied to answer defined retail business questions and transform transactional records into structured performance insights.

8. Tableau Dashboard Development

8.1 Dashboard Design

Tableau Public was used to transform the analysed retail data into an interactive dashboard focused on sales and profitability performance. The dashboard was designed to provide a clear visual overview of commercial performance while enabling comparison across key dimensions of the Superstore business.

The visualisations were structured to make it easier to distinguish between revenue generation and actual profitability, helping identify areas where strong sales performance did not necessarily translate into equally strong profit.

8.2 Dashboard Visualizations

- Sales & Profit Performance: provides an overview of the core commercial measures and helps evaluate the relationship between revenue and profitability.
- Category & Sub-Category Performance: compares product groups to identify areas generating stronger sales and profit as well as areas of weaker profitability.
- Customer Segment Analysis: compares performance across customer segments to identify differences in commercial contribution.
- Performance Trends: examines changes in sales and profitability over time to identify growth patterns and fluctuations.
- Profitability Analysis: supports identification of areas where sales performance may be strong but profit contribution is comparatively weak.

8.3 Dashboard Purpose

The dashboard brings multiple dimensions of retail performance into a single analytical view. Rather than evaluating sales in isolation, it enables sales and profit to be compared across

products, categories, customer segments and time. This provides a more commercially meaningful assessment of performance and helps highlight potential profitability risks and opportunities.

8.4 Interactive Dashboard

Tableau Public

https://public.tableau.com/app/profile/anshi.rana6970/viz/RetailSalesProfitabilityAnalysis_17868962281980/RetailSalesProfitabilityDashboard

9. Key Findings & Comparative Analysis

9.1 Sales vs Profit Performance

The analysis demonstrates a clear distinction between **revenue generation and value creation**. Sales indicate the level of customer spending generated by the business, whereas profit shows how effectively that revenue is converted into financial return. Across the dataset, stronger sales performance did not consistently correspond with proportionally stronger profitability.

This divergence suggests that revenue performance alone can provide an incomplete view of commercial success. Two areas of the business may generate similar sales while producing substantially different profit outcomes because of differences in margins, discounting and product mix.

Performance should therefore be evaluated using sales and profit together. Areas with both strong sales and strong profit represent healthier commercial performance, whereas high-sales but comparatively low-profit areas warrant further investigation into margin erosion and discount behaviour.

9.2 Category & Sub-Category Performance

Category-level analysis provides an overall view of product portfolio performance, but the sub-category comparison reveals greater variation within those broader groups. A category can appear commercially successful in aggregate while containing individual sub-categories that contribute considerably less profit or potentially offset gains generated elsewhere.

Comparing sales with profit at both levels is therefore important. High-revenue sub-categories with strong profitability represent valuable areas of the portfolio, while those generating substantial sales but comparatively weak profit indicate that revenue is being converted into financial return less efficiently.

Category-level results should be used for strategic portfolio assessment, while sub-category analysis should support more targeted decisions. This prevents strong aggregate performance from masking weaker product groups and allows management to prioritise profitable areas while investigating underperformance more precisely.

9.3 Customer Segment Performance

The customer-segment analysis demonstrates that commercial contribution should be assessed using both **scale and profitability**. A segment generating a larger proportion of sales may be important because of its revenue contribution, but this does not automatically make it the strongest segment from a profitability perspective.

Comparing sales and profit across segments provides a more balanced assessment of customer value. Where a segment demonstrates strong performance across both measures, it represents a more commercially attractive source of business. Conversely, a large gap between sales contribution and profit contribution may indicate lower margins, differences in purchasing behaviour or greater exposure to discounted transactions.

Customer strategy should therefore move beyond identifying the segment with the highest sales. Segment performance should be evaluated according to how efficiently customer spending is converted into profit, enabling marketing and retention decisions to focus on commercially valuable growth rather than revenue volume alone.

9.4 Yearly Sales & Profit Trends

Analysing performance over time provides an important comparison between **business growth and profitable growth**. An increase in sales between periods is positive from a revenue perspective, but its commercial quality depends on whether profit grows at a comparable rate.

When sales and profit move upward together, the business is more likely to be translating additional revenue into improved financial performance. However, where sales increase more strongly than profit, the divergence may indicate margin pressure caused by discounting, changes in product mix or increased contribution from lower-margin transactions.

Similarly, fluctuations in profit should be interpreted alongside the corresponding sales movement. A decline in profitability becomes more significant when revenue remains comparatively strong because it suggests that the issue may relate to how sales are being generated rather than simply insufficient demand.

Trend analysis should therefore focus on the relationship between sales growth and profit growth rather than viewing either measure independently. Sustainable retail growth requires improvement in revenue without disproportionately sacrificing profitability.

9.5 Discount & Profitability Analysis

The discount analysis provides an important explanation for why high sales do not necessarily translate into high profit. Discounting can stimulate demand and increase transaction value, but it simultaneously reduces the revenue retained from each sale and can therefore place pressure on margins.

Comparing discount levels with profitability enables the business to distinguish between **productive discounting**, where promotional activity still generates acceptable profit, and potentially **value-destructive discounting**, where sales are achieved at the expense of profitability.

This relationship is particularly important when considered alongside product and sub-category performance. An area generating substantial sales but weak or negative profit should be investigated to determine whether aggressive discounting is contributing to the profitability gap.

The objective should not necessarily be to eliminate discounting, but to evaluate whether the incremental sales generated by a discount justify the reduction in margin. Discount decisions should therefore be assessed using profitability outcomes rather than sales uplift alone.

9.6 Product-Level Productivity

Product-level analysis provides the most granular view of performance and demonstrates how individual products can influence wider category and sub-category results. Aggregate sales figures can conceal significant differences between products, including situations where profitable items compensate for losses generated by weaker ones.

Comparing individual products by sales and profit allows them to be viewed from different commercial perspectives. High-sales/high-profit products represent strong performers, while high-sales/low-profit products require margin investigation. Low-sales/high-profit products may represent opportunities for targeted growth, whereas persistently loss-making products require closer examination.

This level of analysis also helps explain findings observed higher in the product hierarchy. Weak sub-category profitability, for example, may be concentrated among a relatively small number of problematic products rather than reflecting poor performance across the entire sub-category.

Product-level diagnosis allows management to take targeted action on pricing, promotion and assortment rather than making broad changes to otherwise profitable categories.

9.7 Cross-Dimensional Performance Analysis

Examining the analyses together shows that profitability is influenced by interconnected factors rather than a single business dimension. Category and sub-category performance identifies **where** profit is generated, segment analysis identifies **which customer groups** contribute to performance, trend analysis establishes **when** performance changes, and discount and product analysis help explain **why** profitability may strengthen or deteriorate.

For example, high sales within a category should not automatically be interpreted as strong performance if sub-category or product analysis reveals weak profitability. Similarly, strong revenue from a customer segment becomes less commercially attractive if it depends heavily on discounted, low-margin transactions.

The strongest business decisions therefore come from combining these dimensions rather than ranking performance using one KPI. This enables management to distinguish between genuine profitable growth and revenue growth that may be masking underlying margin weakness.

9.8 Overall Analyst Insight

The central finding of the analysis is that **sales growth and profitable growth are not the same outcome**. Revenue identifies where demand exists, but profitability determines whether that demand is creating sustainable commercial value.

The combined evidence suggests that retail performance should be evaluated through a hierarchy of analysis: overall sales and profit establish business performance; categories and segments identify where performance originates; sub-categories and products expose areas of strength or weakness; and discount analysis helps explain potential margin pressure.

The priority should therefore be to protect and expand areas demonstrating both strong demand and healthy profitability, while investigating high-revenue areas where profit contribution is disproportionately weak. This creates a more commercially meaningful approach to product, customer and pricing strategy than maximising sales alone.

10. Recommendations

10.1 Prioritize Profitable Growth

Management should evaluate sales and profit together rather than using revenue growth as the primary measure of success. Areas demonstrating both strong sales and healthy profitability should receive greater strategic attention, while high-sales but comparatively low-profit areas should be investigated for margin pressure.

10.2 Address Underperforming Sub-Categories

Sub-categories with weaker profitability should be reviewed independently rather than making broad changes at category level. Management should investigate product mix, pricing and discount behaviour within these areas to determine the specific drivers of underperformance.

10.3 Optimize Discount Strategy

Discounting should be evaluated according to its effect on profit rather than sales uplift alone. Higher discount levels should be reviewed to determine whether the additional demand generated is sufficient to compensate for reduced margins, particularly for products already demonstrating weak profitability.

10.4 Focus Customer Strategy on Profitable Segments

Customer segments should be compared using both revenue contribution and profitability. Marketing and retention activity can then be prioritised toward segments that demonstrate stronger commercial value, while lower-margin segments can be assessed for opportunities to improve profitability.

10.5 Review Loss-Making Products Individually

Persistently loss-making products should be investigated at product level to identify whether pricing, discounting or other product-specific factors are contributing to negative profitability. Targeted intervention is preferable to making changes across an entire category containing otherwise profitable products.

10.6 Monitor Sales and Profit Trends Together

Sales growth should be monitored alongside profit growth to identify potential margin deterioration early. If revenue increases without a comparable improvement in profit, discount levels, product mix and the sources of sales growth should be investigated before concluding that overall performance has improved.

10.7 Develop the Dashboard as a Retail Performance Tool

The dashboard could be updated with new transactional data and expanded with measures such as profit margin, average order value and customer profitability. This would enable management to monitor performance continuously, identify emerging profitability risks and support more informed product, pricing and customer decisions.

Overall, the analysis supports a shift from a sales-focused view of performance toward a profitability-led approach. Combining sales, profit, discount, product and customer

information enables the business to identify where genuine value is being created and where revenue growth may be masking weaker commercial outcomes.

11. Skills Demonstrated

- Data Cleaning & Preparation: Reviewed, structured and prepared transactional retail data in Google Sheets for analysis.
- SQL & BigQuery: Used SQL aggregation, filtering, grouping and ordering techniques to analyse sales, profitability, customer segments, products and discounts.
- Data Visualisation: Developed an interactive Tableau dashboard to communicate retail sales and profitability performance.
- Commercial Performance Analysis: Compared sales against profit to distinguish revenue generation from profitable growth.
- Trend & Comparative Analysis: Evaluated performance across categories, sub-categories, customer segments, products and time.
- Profitability Analysis: Investigated discount behaviour and product-level performance to identify potential sources of margin pressure.
- Business Analysis: Translated transactional data into commercially relevant findings and recommendations.
- Data Storytelling: Connected KPIs, trends and detailed performance analysis into a clear decision-focused narrative.

12. Conclusion

This project demonstrates an end-to-end business analytics workflow for evaluating retail sales and profitability using transactional Superstore data. The analysis shows that revenue generation alone does not provide a complete assessment of commercial performance, as profitability varies across categories, sub-categories, customer segments, products and discount levels.

By combining Google Sheets, SQL in Google BigQuery and Tableau, the project transformed transactional data into structured performance analysis and business-focused insights. The findings reinforce the importance of evaluating sales alongside profitability and using more granular analysis to identify where commercial value is being created, where margins may be under pressure and where targeted action may improve performance.

13. Project Links

Interactive Tableau Dashboard:

Retail Sales & Profitability Analysis:

https://public.tableau.com/app/profile/anshi.rana6970/viz/RetailSalesProfitabilityAnalysis_17868962281980/RetailSalesProfitabilityDashboard

Sample Superstore Retail Dataset:

<https://www.kaggle.com/datasets/divyjn28/superstore-dataset>